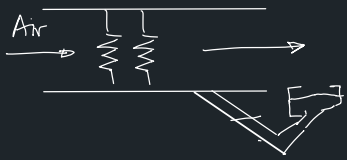




10/7 CH9 Measurement of Pressure and velocity

In lab "Inclined manometer"

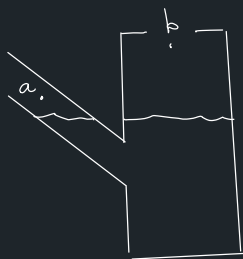


Absolute pressure

$$P_{abs} = \frac{\text{Force}}{\text{Area}}; P_{abs} < 0$$

$$P_{gauge} = P_{abs} - P_{atm}; P_{gauge} \in \mathbb{R}$$

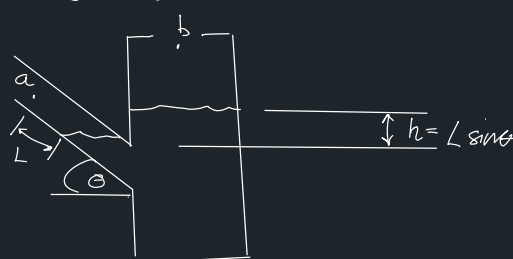
State I



$$P_{gauge} = 0$$

$$P_a = P_b$$

State II

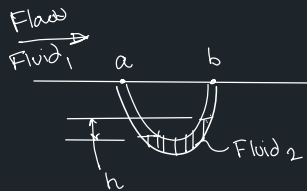


$$P_{gauge} > 0$$

$$P_a > P_b$$

Manometers: measures pressure differences

Fluid statics: $P_a - P_b = gh(\rho_1 - \rho_2)$



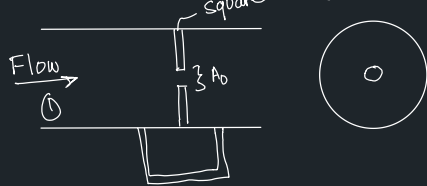
CH 10 Flow measurement - Measure $\dot{m}$ or $Q = \frac{\dot{m}}{\rho}$

Obstruction Meter

- Meter acts as an obstruction to the flow
- This causes pressure changes which are related

Types:

- (1) Venturi
- (2) Flow nozzle
- (3) Orifice flow meter



$$Q_1 = K A_0 \sqrt{\frac{2\Delta P}{\rho_1}} Y$$

$$Q_1 = \text{upstream volumetric flow rate} \left[\frac{\text{m}^3}{\text{s}} \right]$$

A_0 = Minimum flow area

$$\Delta P = P_a - P_b$$

$$\rho_1 = \text{upstream density} \left[\frac{\text{kg}}{\text{m}^3} \right]$$

$$Y = \text{expansion factor}$$

$$K_0 = \text{flow coefficient}$$

Example $\Delta P = 21.5 \text{ kPa}$

$T_1 = 20^\circ\text{C}$ Square faced orifice
 $P_1 = 93.7 \text{ kPa (abs)}$
 $Q_1 = ?$

Use ideal gas law to find ρ_1 : $PV = mRT$

$$\frac{P}{P_{air}T} = \frac{m}{V} = \rho$$

$$\text{Where } P_{air} = 283 \frac{\text{J}}{\text{kgK}}$$

$$T_1 = 293 \text{ K}$$

$$P_1 = 93700 \text{ Pa}$$

$$A_0 = \frac{\pi}{4} d_0^2 = 4.52 (10^{-4}) \text{ m}^2; d_0 = 0.024 \text{ m}$$

Chart for Y vs K_0

$$\text{Wherein } K_0(R_{e_{d_1}}); R_{e_{d_1}} = \frac{4Q_1 \rho_1}{\pi d_1 \mu_1}$$

10/9 Lab report format

for case 1 in lab report, P_1 (gauge)

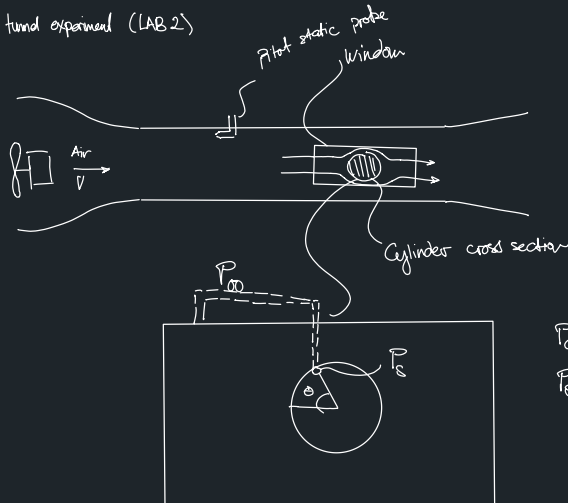
$$P_{abs} = P_1 (\text{gauge}) + P_{atm} (\text{during lab})$$

from WB, DB, CPD

$$\dot{m}(h_D - h_C) = \dot{Q}_{in}$$

$$\dot{m} = \frac{\dot{Q}_{in}}{h_D - h_C}$$

Wind tunnel experiment (LAB 2)



P_0 : static upstream pressure

P_s : surface pressure